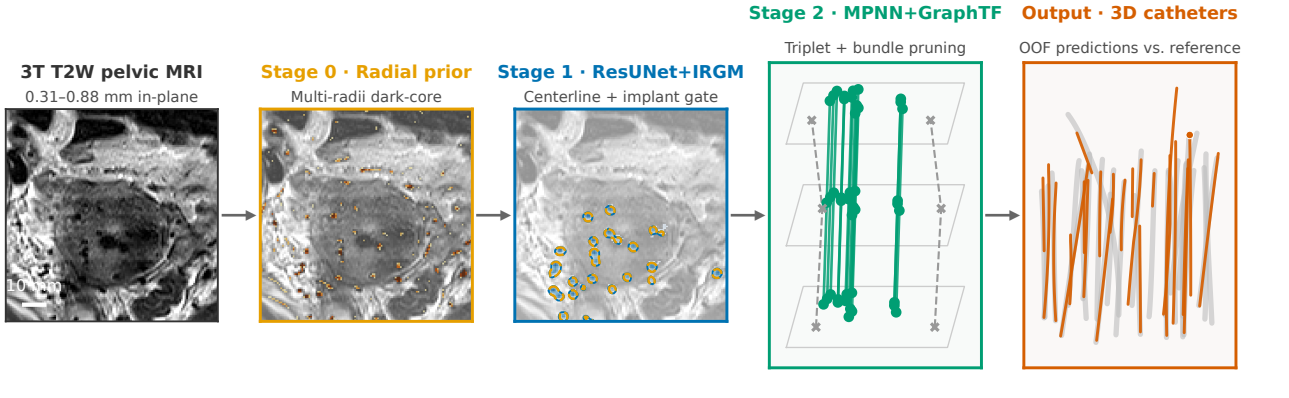
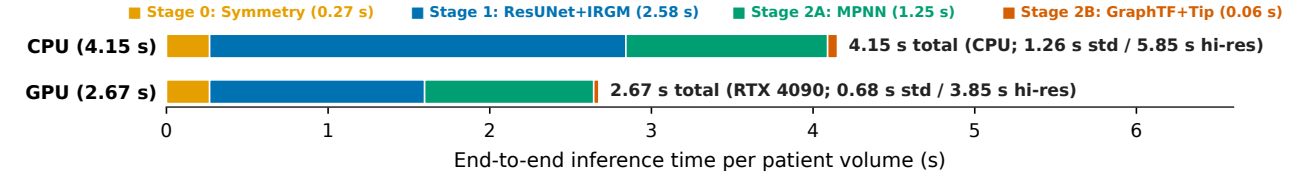

3T T2W pelvic MRI
0.31–0.88 mm in-plane
10 mm
Stage 0 · Radial prior
Multi-radii dark-core
Stage 1 · ResUNet+IRGM
Centerline + implant gate
Stage 2 · MPNN+GraphTF
Triplet + bundle pruning
Output · 3D catheters
OOF predictions vs. reference


Stage 0: Symmetry (0.27 s)
Stage 1: ResUNet+IRGM (2.58 s)
Stage 2A: MPNN (1.25 s)
Stage 2B: GraphTF+Tip (0.06 s)
CPU (4.15 s)
4.15 s total (CPU; 1.26 s std / 5.85 s hi-res)
GPU (2.67 s)
2.67 s total (RTX 4090; 0.68 s std / 3.85 s hi-res)
0
1
2
3
4
5
6
End-to-end inference time per patient volume (s)